

1. FULL-STACK TECHNOLOGY ARCHITECTURE

- **Frontend Engine:** Next.js 16 (App Router + Turbopack) + React 19 for instant hydration and sub-50ms render times.
- **Language & Types:** 100% Strict TypeScript 5 with unified type contracts for IoT events, alerts, and surplus items.
- **Database Layer:** MongoDB Atlas (17+ synchronized collections) with aggregation pipelines for real-time querying.
- **UI & Visuals:** Tailwind CSS v4 + Framer Motion animations + Recharts for hourly telemetry and spoilage curves.

2. KITCHEN MESS: AI DEMAND PREDICTION & HUMAN-IN-THE-LOOP

- **Core Algorithm:** Multi-variate regression model ingesting: Day-of-Week, 14-day rolling average, weather & exams.
- **Dynamic Buffer:** Automated buffer reduction: Throttles traditional 15% kitchen safety buffer down to 4.2%.
- **Human-in-the-Loop:** Manager Override slider with reason logging. Model feedback loop learns from manager inputs.
- **Impact Achieved:** 32.4% reduction in mess food waste, saving Rs. 1.8 Lakh monthly in raw kitchen procurement.

3. INDUSTRIAL FACTORY: WEIBULL SPOILAGE DECAY & IOT TELEMETRY

- **Spoilage Math:** Non-linear Weibull Hazard Function: $Q(t) = Q_0 * \exp(-(t / \eta)^\beta)$. Models accelerated spoilage.
- **Multi-Sensor Input:** Continuously ingests cold storage temperature, relative humidity, and Ethylene Gas (PPM).
- **Anomaly Detection:** Optical caliper telemetry (PM-03 drum) checks peel thickness drift (>1.2mm); calculates excess loss rate.
- **Job Prioritization:** One-click 'Prioritize Batch' re-sequences production queue to Front-of-Line, salvaging 2,800 kg.

4. LOGISTICS: VRPTW OPTIMIZATION & FSSAI SURPLUS PROTOCOL

- **Routing Problem:** Solves Vehicle Routing Problem with Time Windows (VRPTW) using Dijkstra/Nearest Neighbor graph.
- **2-Hour Safety Lock:** Enforces strict 120-minute delivery window under Indian Surplus Food Regulations, 2019.
- **Food Quality Gate:** Requires digital holding temperature confirmation (>65°C hot holding, <5°C cold chain).
- **Secure Handshake:** 6-digit OTP verification + tamper-evident packaging checklist before NGO driver custody release.

5. ESG CARBON ACCOUNTING & CLIENT-SIDE PDF GENERATOR

- **Carbon & Water Math:** 1 kg food rescued = 2.5 kg CO2e saved; 1 cooked meal rescued = 140 Litres virtual water conserved.
- **Gamification Tiers:** Bronze (<500), Silver (500-1999), Gold (2000-4999), Platinum (5000+) based on NGO feedback.
- **Vector PDF Engine:** Base64 vector pipeline embedding official FoodWise Logo, FSSAI Certified Seal & 'Every Meal Counts'.
- **Tamper-Proof Ref:** Cryptographic hash generation (e.g. FW-DONOR-2002840) verifiable against MongoDB audit stream.